

UDCT v7.09 — Pre-registration

Applying the v7.06 virial and Jeans protocol to Milgrom's published GQUMOND length-screening example, with a QUMOND control, in three ultra-faint dwarfs

30 September 2026 (revised before deposit: adds effective-mass diagnostics) | Won Shik Paik, independent researcher, Auckland, New Zealand | for deposit before any calculation

Declaration. At deposit, no calculation of the theory below has been made: no virial integral, no Jeans pressure, no dispersion. The objects, profiles and Galactic field are those already used in v7.05–v7.06, whose results for a different (UDCT) action are known to the author; the results for the theory registered here are not. The question is whether the negative virial work found in v7.06 is a property of Hessian-dependent GQUMOND theories in general or of the UDCT integral construction (v7.08).

1. Theory under test (frozen)

Milgrom (2023), GQUMOND example with screening of small systems [1, Eqs. 13, 15, 16]:

$$L_G = -(1/8\pi G)[2\nabla\phi\cdot\nabla\psi - a_0^2 P], \quad P = f(u) \, Q(Z/f), \quad Z = (\nabla\psi)^2/a_0^2, \quad u = (\nabla\psi)^2/(\ell_0^2 \psi_{,ij} \psi_{,ij}), \quad f(u) = u/(1+u),$$
$$\nabla^2\phi = (a_0^2/2)[\nabla_i(\partial P/\partial\psi_{,i}) - \nabla_i\nabla_j(\partial P/\partial\psi_{,ij})] \text{ (full source, both terms kept; [1], Eq. 7; reduces to } \nabla\cdot(Q'\nabla\psi) \text{ for } P = Q).$$

Choice	Registered value
Q	the QUMOND function with $Q'(Y^2) = v_0(Y) = (1 + 1/Y)^{1/2}$ (the SPARC-locked law of the campaign), $Q(0) = 0$
a_0	$1.082401 \times 10^{-10} \text{ m s}^{-2}$
ℓ_0 grid	0.03, 0.1, 0.3, 1, 3, 10 pc. Lower edge: $\approx 6000 \text{ AU}$, the solar scale Milgrom names for suppressing the Galactic anomaly [1]; upper edge well below galactic scales, as he requires
Control	QUMOND itself ($f \equiv 1$, i.e. $\ell_0 \rightarrow 0$), same objects and inputs: this is also the like-for-like control F5 requested in v7.06
f	exactly $u/(1+u)$; no other screening function is tried under this registration

2. Objects, environment and numerics (frozen, inherited from v7.06)

Leo IV, Pegasus III, Hydrus I with the v7.05 inputs (L, R_e from the Local Volume Database [4]; $M_*/L = 2$); Plummer ($b = R_e$) and tapered Hernquist-like cusp profiles; Galaxy as the local second-order Taylor field (acceleration E and Hessian H_{MW}) of the visible McMillan (2017) disks plus an $8.62 \times 10^9 M_\odot$ point bulge [3]; no halo, no LMC. Implementation: the released v7.06 script (SHA-256 prefix a64440e1350ae8bf) with only the constitutive law replaced; $\partial P/\partial\psi_{,i}$ and $\partial P/\partial\psi_{,ij}$ evaluated analytically; angular and radial grids at the v7.06 defaults (360 radial; 24×36 directions), with the v7.06 smoke grid as a convergence check. Sign conventions, the integration-by-parts evaluation of V and the zero-outer-pressure Jeans integral are unchanged.

3. Quantities and decision rules (fixed now)

Quantity	Registered use
V/V_N for each object $\times$ profile $\times \ell_0$ (36 cases) and for the QUMOND control (6 cases)	primary
share of the Hessian-derivative term in V	diagnostic
angle-averaged radial force $\langle g_r \rangle/g_*$ at r_h and its sign over $0.1\text{--}10 r_h$; enclosed effective mass $M_{\text{eff}}(<r) = M_*(<r) + M_{\text{phantom}}(<r)$, obtained from the Gauss-law identity $M_{\text{eff}}(<r) = r^2 \langle g_r \rangle / G$ (inward-positive $\langle g_r \rangle$, so a negative M_{eff} corresponds to an outward mean force); the smallest value of $M_{\text{eff}}(<r)$ and whether it is negative	diagnostic, reported for every case and for the QUMOND control; the question addressed is whether any of these theories makes the effective mass inside the stellar body negative under a strong external field
minimum σ_r^2 (isotropic Jeans) and projected LOS rms inside R_e	secondary; archived dispersions used as a scale reference only
convergence	a sign is claimed only if it agrees on the default and smoke grids

Two clarifications are fixed now. (a) A negative M_{eff} or outward mean force is reported as an *equilibrium diagnostic in a static model*, not as a prediction that stars are ejected; no time evolution is computed. (b) A magnitude of the outward force exceeding the stellar Newtonian force by more than a factor 10 is flagged as a warning of a construction or approximation problem (for example the

uniform-external-field Taylor expansion extended to tens of r_h), not as a physical effect, and is followed by a check with the Taylor field truncated at $5 r_h$. **Outcome A — obstruction shared:** $V/V_N < 0$ in at least one object and profile at some ℓ_0 in the grid while the QUMOND control is positive. Reported as "Milgrom's example A exhibits negative virial work in the stated models for ℓ_0 in [range]". **Outcome B — construction-specific:** $V/V_N > 0$ for all 36 cases. Reported as "the v7.06 obstruction does not appear in this GQUMOND example; it is attributed to the UDCT integral construction". **Outcome C — control negative:** if QUMOND itself gives $V/V_N < 0$ anywhere, the protocol or inputs are suspect and no claim about GQUMOND is made until resolved. Jeans results are reported under every outcome but do not change A/B/C.

4. Limits declared in advance

(i) Only one GQUMOND example and one f ; other members of the class are not addressed. (ii) The Galaxy is a local Taylor field; no global nonaxisymmetric solve. (iii) Spherical isotropic Jeans reduction, no anisotropy, binaries, tides or time dependence. (iv) Three objects only; no statistical claim. (v) Milgrom's Q normalization differs in convention from the campaign's v_0 ; the registered Q is the campaign's choice, stated above. (vi) Passing or failing does not bear on UDCT, which is not the theory under test.

5. Disclosure of assistance

Design of this registration and the planned calculation are carried out with the assistance of large language models (Anthropic Claude; earlier v7.xx work also used OpenAI ChatGPT, Google Gemini and xAI Grok) under the author's direction. The code will be released with the results for independent reproduction.

References

- [1] M. Milgrom, Generalizations of quasilinear MOND, Phys. Rev. D 108, 084005 (2023), arXiv:2305.01589.
- [2] M. Milgrom, Quasi-linear formulation of MOND, MNRAS 403, 886 (2010), doi:10.1111/j.1365-2966.2009.16184.x.
- [3] P. J. McMillan, The mass distribution and gravitational potential of the Milky Way, MNRAS 465, 76 (2017), doi:10.1093/mnras/stw2759.
- [4] A. B. Pace, Local Volume Database (2025), doi:10.33232/001c.144859.
- [5] UDCT v7.06 research record and released script (29 September 2026); v7.08 prior-art correction (30 September 2026).